MP5031

USB PD source controller

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Introduction

The main purpose of this library is to simplify communication process between host MCU and slave device. Communication is handled by I^{2}C protocol. Please note that, this library has been created as hobby project and was not meant to be ideal.

MP5031 USB PD source controller was designed by Monolithic Power Systems. This IC is simple to use and relatively cheap device, that allows designers to easily integrate PD capabilities in their designs via I^{2}C protocol.

This USB PD Port Controller with Five Configurable PDOs and Load-Shedding option, can be integrated with I^{2}C power converter IC, such as MP4245, MP4247 and MP4255. Integration between components is also done by I^{2}C protocol.

MP5031 also features other dedicated charging port schemes, such as QC3.0, FCP and BC1.2. More information can be found in manufacturer datasheet.

Library components

Constructor

Library has been based on OOP (Object Oriented Programming), therefore before usage of any component, user has to create an object with following scheme:

MP5031 nameOfObject (STD_ADR);

MP5031 is the name of used device.

nameOfObject is the name of created object, that user will refer to - for example **myPD.**

STD_ADR is the standard I^{2}C address of the device, which is **0x28.**

getSoftwareRevision()

Library allows the user to check what software revision is being utilised in device. This function will return **int** type value. Function does not utilise additional input arguments.

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setPowerDataObject(int profile, int mode)

This function allows user to enable specified Power Data Object (PDO). First argument decides which PDO will be addressed. Available keywords for this are: **PDO2, PDO3, PDO4 and PDO5.** It is important to note, that PDO1 will always be enabled, because it is the standard 5V PDO. Second argument decides if PDO is enabled or not. User can use keywords **ENABLE** and **DISABLE** in order to set proper value.

setPowerDataObjectMode(int profile, int mode)

This function allows user to set Power Data Object as fixed or augmented. Main difference betweeen those two modes, is that Fixed PDO, or FPDO, can change output voltage in specified voltages that are 5V, 9V, 12, 15V and 20V. Augmented Power Data Object, or APDO, on the other hand can be programmed dynamically with 20mV steps. In order to set proper PDO profile, user has to firstly address which PDO will be selected via first argument of the function. Available keywords for this are: **PDO2, PDO3, PDO4 and PDO5.** Second variable decides what type of PDO will be set. In order to set proper profile, user can use keywords **FPDO** or **APDO**.

setPowerDataObjectModeCurrent(int profile, int current)

This function allows user to set maximum output current of selected PDO. 10mA steps are for FPDO object and 50mA steps are for APDO object. First argument decides which PDO will be addressed. Available keywords for this are: **PDO1, PDO2, PDO3, PDO4 and PDO5.** Because of USD PD specification, second argument can take values ranging from **0 to 5000,** which corresponds to **0A to 5A.**

setPowerDataObjectModeVoltage(int profile, int minVoltage, int maxVoltage)

This function allows user to set maximum output current of selected PDO. 50mV steps are for FPDO object and 100mV steps are for APDO object. First argument decides which PDO will be addressed. Available keywords for this are: **PDO2, PDO3, PDO4 and PDO5**. Value of second argument depend on what type of object is addressed. In the case of FPDO, second argument is the value of chosen PDO voltage. Because of USD PD specification, second argument can take values ranging from **0 to 20000,** which corresponds to **0V to 20V. Third argument is not used in this case.** In the case of APDO, second argument is the minimal required voltage of APDO object. Third argument sets maximal required voltage of APDO object. Because of USD PD specification, second argument can take values ranging from **0 to 21000,** which corresponds to **0V to 21V.**

getSuffixCode()

This function returns so called suffix code. This 4 digit number is set by MPS. Please refer to datasheet for further information.

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setLowVoltageDetectionResponse(int response)

If the GPIO2 or GPIO4 are set to sense input voltage of battery the pack, that powers the device, this function allows user to set proper behaviour, if these GPIOs become active. There are three available options that user can use: **NONE, PDO1 and PWR_SHARE_LIMIT.** Please refer to datasheet p.20 for further information.

setLegacyChargingMode(int mode)

This function allows user to chose what lagacy charging modes will be enabled. The MP5031 supports 5 different legacy charging modes: QC3.0, Huawei FCP, BC1.2 DCP, and Apple divider 3 mode. Available keywords for setting legacy charging modes are: **ALL, DCP and DIVIDER.** Please refer to datasheet p.30 for further information.

setNTCPowerSharingFunction(int mode)

This function enables the NTC2 power share function. For further information, please refer to datasheet p.22. This function accepts keywords: **DISABLE and ENABLE.**

setChargingDataPortMode(int mode)

This function enables the charging data port (CDP) mode. This feature normally is set as disabled, which means charge only protocol is allowed on DP and DM pins. If CDP mode is enabled, QC3.0 mode, Apple divider 3 mode, and DCP mode are disabled. For more information please refer to datasheet p.22. This function accepts keywords: **DISABLE and ENABLE.**

setSlaveAddress(int address)

This function allows user to set I^2C address of the device in slave mode. Normally, this address is set to **0x28.** Address can be changed to the value ranging from **0x2F to 0x20.** This function accepts addresses given within above range.

setTypeCMode(int current)

This function allows user to set 3A or 1.5A Type C mode. In order to set 3A mode, user must provide as an argument number **3** and for 1.5A mode, providing **1** as an argument us required.

setSlaveDevice(int address)

The MP5031 is specifically designed to work with power converters designed by MPS such as **MP298X, MP424X, MP28167, MP42550 and MP42551.** Mentioned devices can be treated as function argument. Additionally, user can provide **DIFF** keyword for non – I^2C power converter. For more information about this topic, please refer to datasheet p.31.

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reset()

This function sends reset command to USB DP sink device. Function does not utilise additional input arguments.

update()

This function sends update command about available PDO profiles to sink USB DP sink device. User has to call this function manually, after changing PDO profile values in order for them to take effect. Function does not utilise additional input arguments.

setUSBSuspend(int mode)

This function sets whether USB suspend functions are supported. This bit is defined in the Fixed Supply PDO message bit 28. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.** For more information, please refer to USB PD specification.

setUSBCommunication(int mode)

This function sets whether USB communication capabilities are supported. This bit is defined in the Fixed Supply PDO message bit 26. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.** For more information, please refer to USB PD specification.

setLPSCompliance(int mode)

This function sets the LPS compliance. These bits are defined in the **SOURCE_CAPABILITIES_EXTENDED**. Function utilises arguments in the form of specified keywords: **LPS, PS1 and PS2.** For more information, please refer to USB PD specification.

setTouchCurrent(int mode)

This function sets the value of touch current. These bits are defined in the **SOURCE_CAPABILITIES_EXTENDED.** Function utilises arguments in the form of specified keywords: **EPS, GP and GPE.** For more information, please refer to USB PD specification.

setTouchTemperature(int mode)

This function sets the value of touch current. These bits are defined in the **SOURCE_CAPABILITIES_EXTENDED**. Function utilises arguments in the form of specified keywords: **TS0, TS1 and TS2**. For more information, please refer to USB PD specification.

setChunkedBit(int mode)

This function sets the unchunked extend message bit. This bit is defined in Fixed Supply PDO message bit 24. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.** For more information, please refer to USB PD specification.

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setAmountOfSentSourceCapability(int time)

This function allows user to set the amount of source capability bits being send to PD2.0 device, when attached. Function utilises an argument in the form of keyword: **ONCE and TWICE.** For more information, please refer to USB PD specification.

setVDRV(int mode)

This function allows user to enable external N-channel MOSFET gate driver signal. If the sink is attached, then the VDRV pin turns the external N-channel MOSFET on, and power flows from the bus to the sink. If the sink is detached, VDRV turns the external Nchannel MOSFET off to isolate the power path. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.**

setOperationMode(int mode)

This function allows user to choose what type of current modulation (PFM or PWM) will be used, while using MP298X or MP28167 – A power converters. Function utilises an argument in the form of keyword: **PFM and PWM.** For more information about this topic, please refer to MP298X or MP28167 datasheets.

setFrequencyDithering(int mode)

This function allows user to set dithering function on MP298X devices. This feature allows the power converter to lower peak EMI signature in order to pass strict compliance standards. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.** For more information, please refer to MP298X datasheet.

setTimerDelay(int mode)

In the case of using GPIO4 as enable pin, device off timer can be set. If GPIO4 pin drops below 1.2V, MP5031 enters low I_Q state. Delay of this entrance can be set via this function. Function utilises an argument in the form of delay values in minutes: **0, 10, 22, 40, 80 and 120.**

setGPIO4Functionality(int mode)

This function allows user to set GPIO4 functionality. Possible options include using GPIO4 as: not used, enable, NTC2 and VBATT sense, In order to set proper behaviour on this GPIO, user can use following keywords: **NONE, EN, NTC2 and VBATT.** For more information about this topic, please refer to MP5031 p.33 datasheet.

setGPIO3Functionality(int mode)

This function allows user to set GPIO3 functionality. Possible options include using GPIO3 as: I^2C arbitrary, enable, attach, enable delay timer, polarisation, PDO3 selection, I_{SENSE+} and I_{PWM}. In order to set proper behaviour on this GPIO, user can use following keywords: **ARB, EN,**

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ATTACH, EN_OFF, POL, PDO3_SEL, ISENSP and IPWM. For more information about this topic, please refer to MP5031 p.33 datasheet.

setGPIO2Functionality(int mode)

This function allows user to set GPIO2 functionality. Possible options include using GPIO2 as: 12V Quick Charge, polarisation, pulled low, power share function and VBATT sense. In order to set proper behaviour on this GPIO, user can use following keywords: **QC, POL, PULLDOWN, SHARE and VBATT.** For more information about this topic, please refer to MP5031 p.33 datasheet.

setGPIO1Functionality(int mode)

This function allows user to set GPIO1 functionality. Possible options include using GPIO1 as: PDO2 selection, interrupt, pulldown, open drain and VSEL1. In order to set proper behaviour on this GPIO, user can use following keywords: **PDO2_SEL, INT, PULLDOWN, OD and VSEL1.** For more information about this topic, please refer to MP5031 p.34 datasheet.

setNTCBehaviour(int mode)

This function allows user to set device behaviour, in case of over temperature scenario. There are 2 possible modes: Load – shedding and shut down. If load shedding is chosen, the device will limit the power that is drawn from power supply (converter). If shut down is chosen, the device will engage thermal shutdown. Function utilises an argument in the form of special keyword: **REDUCE and SHUTDOWN.**

setUVLOThreshold(int mode)

This function allows user to set VBUS_P UVLO threshold level. If the minimum APDO voltage is 3.3V, then set the UVLO threshold to 2.97V. If the minimum APDO voltage is 5V, then set the UVLO threshold to 4.5V. Function utilises an argument in the form of threshold voltage which can be: **3 or 5**

setVDRVFunctonality(int mode)

This function allows user to set functionality of VDRV feature. This functionality can be set via function argument set either to: **VDRV0 or VDRV1.** For more information about this topic, please refer to MP5031 p.19 datasheet.

setGPIO7Functionality(int mode)

This function allows user to set GPIO7 functionality. Possible options include using GPIO7 as: discharge, attach, enable out mid, enable out and adjust., In order to set proper behaviour on this GPIO, user can use following keywords: **DISCHG, EN_OUT_MID, EN_OUT and COMP.** For more information about this topic, please refer to MP5031 p.33 datasheet.

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setGPIO6Functionality(int mode)

This function allows user to set GPIO6 functionality. Possible options include using GPIO6 as: PDO4 selection, VSEL2, polarisation, enable, sync out, NTC2 and I^2C slave SDA pin. In order to set proper behaviour on this GPIO, user can use following keywords: **PDO4_SEL, VSEL2, POL, EN, SYNC_OUT2, and I2C_SLV_SDA.** For more information about this topic, please refer to MP5031 p.34 datasheet.

setGPIO5Functionality(int mode)

This function allows user to set GPIO5 functionality. Possible options include using GPIO5 as: I^2C slave SCL pin, IPWM, synchronisation pin, attach and VBUS UV. In order to set proper behaviour on this GPIO, user can use following keywords: **I2C_SLV_SDA, IPWM, SYNC_OUT1, ATTACH and VBUS_DETECT**. For more information about this topic, please refer to MP5031 p.33 datasheet.

setSinkPowerDataObjectPowerRating(int value)

This function allows user to set sink – requested PDO power rating threshold. The default value is 0x3C (60W). Function argument accepts values ranging from **0 to 255**, which represents power rating in Watts.

setSoucePowerDataObjectPowerRating(int value)

This function allows user to set port's PD power rating. All PDOs with a power rating that meets or exceeds this threshold are disabled. The default value is 0x3C (60W). Function argument accepts values ranging from **0 to 255**, which represents power rating in Watts.

getSinkRequestedVoltage()

This function returns voltage requested for PPS in 20mV steps. This bit is only valid when the selected PDO is APDO. Value is returned as **int** type.

checkDataLineStatus()

This function returns logic high, if CC1, CC2, DP, or DM is shorted to the input battery. Value is returned as **int** type.

checkAttachStatus()

This function returns logic high whether a Type - C device is attached. Value is returned as **int** type.

checkOverTemperatureEvent()

This function returns logic high whether the MP5031 has entered over - temperature protection (OTP), excluding thermal shutdowns caused by NTC or NTC2. It also indicates if the external power device enters OTP. Value is returned as **int** type.

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checkContract()

This function returns logic high whether the MP5031 has a PD contract. Value is returned as **int** type.

checkLoadShedding()

This function returns logic high whether the device has entered load – shedding state. Value is returned as **int** type.

checkCableCapability()

This function returns current capability of attached cable. Function returns **int** type value, which can be equal to 3 or 5, corresponding to 3A and 5A.

checkMismatchEvent()

This function returns logic high, if sink request sets the capability mismatch bit. Value is returned as **int** type.

getSinkPowerDataObjectDemand()

This function returns sink - requested PDO position, if a sink is attached. Function returns **int** type value, which value can be ranging from 0 to 5, where 0 is non - PD device is attached.

getSinkRequestedCurrent()

This function returns returns the sink-requested APDO current in 50mA increments. Function returns **int** type value.

getVendorIdentificationNumber()

This function returns vendor ID. The default is 8. Function returns **int** type value.

setDigitalClock(mode)

This function allows the user to enable or disable digital clock, which enables or disables device autonomous mode. Function utilises an argument in the form of keyword: **DISABLE and ENABLE.** This function has to be called, after configuring device.

Keywords

User can use following keywords in designed sketch:

Power Data Object: PDO1, PDO2, PDO3, PDO4, PDO5, APDO, FPDO
Permission: ENABLE, DISABLE, NONE, ALL
Legacy charging: DCP, DIVIDER

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Slave devices: MP298X, MP424X, MP28167, MP42550, MP42551, DIFF
USB PD specification: LPS, PS1, PS2, EPS, GP, GPE, TS0. TS1, TS2
Repeat times: ONCE, TWICE
Power converter operation: PFM, PWM
NTC mode: REDUCE, SHUTDOWN
VDRV mode: VDRV0, VDRV1
GPIO1: PDO2_SEL, INT, PULLDOWN, OD, VSEL1
GPIO2: QC, POL, PULLDOWN, SHARE, VBATT
GPIO3: ARB, EN, ATTACH, EN_OFF, POL, PDO3_SEL, ISENSP, IPWM
GPIO4: NONE, EN, NTC2, VBATT
GPIO5: I2C_SLV_SDA, IPWM, SYNC_OUT1, ATTACH, VBUS_DETECT
GPIO6: PDO4_SEL, VSEL2, POL, EN, SYNC_OUT2, I2C_SLV_SDA
GPIO7: DISCHG, EN_OUT_MID, EN_OUT, COMP

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